

TABLE 4
RUNTIME WITH 2.7V 10F SUPERCAPACITOR

Scenario	Total Avg. Current	V _{final}	ΔV	Runtime with 3F (single)	Runtime with 10F (this)	Approx. Improvement vs 3F	Notes
Sleep/standby (OLED off + MCU power-down)	~10μA	1.8V	0.9V	~3.1 days	~10-11 days	×3.3	Ultra-low consumption; ideal for infrequent wake-ups
Idle display (OLED black screen + MCU idle)	~2mA	1.8V	0.9V	~22.5 minutes	~74 minutes (~1 hr 14 min)	×3.3	Display ON but minimal activity
Continuous run (OLED typical display + MCU typical)	~13mA	2.7V	~0.1V	~23 seconds	~77 seconds	×3.3	Very short runtime; voltage quickly drops below 2.7V
Continuous run (OLED typical + MCU active)	~13mA	1.8V	0.9V	~3 min 28 sec	~11 min 30 sec	×3.3	Practical continuous runtime; clock reduction may be required below 2.7V
Max load (OLED full white + MCU active)	~28mA	1.8V	0.9V	~1 min 36 sec	~5 min 20 sec	×3.3	

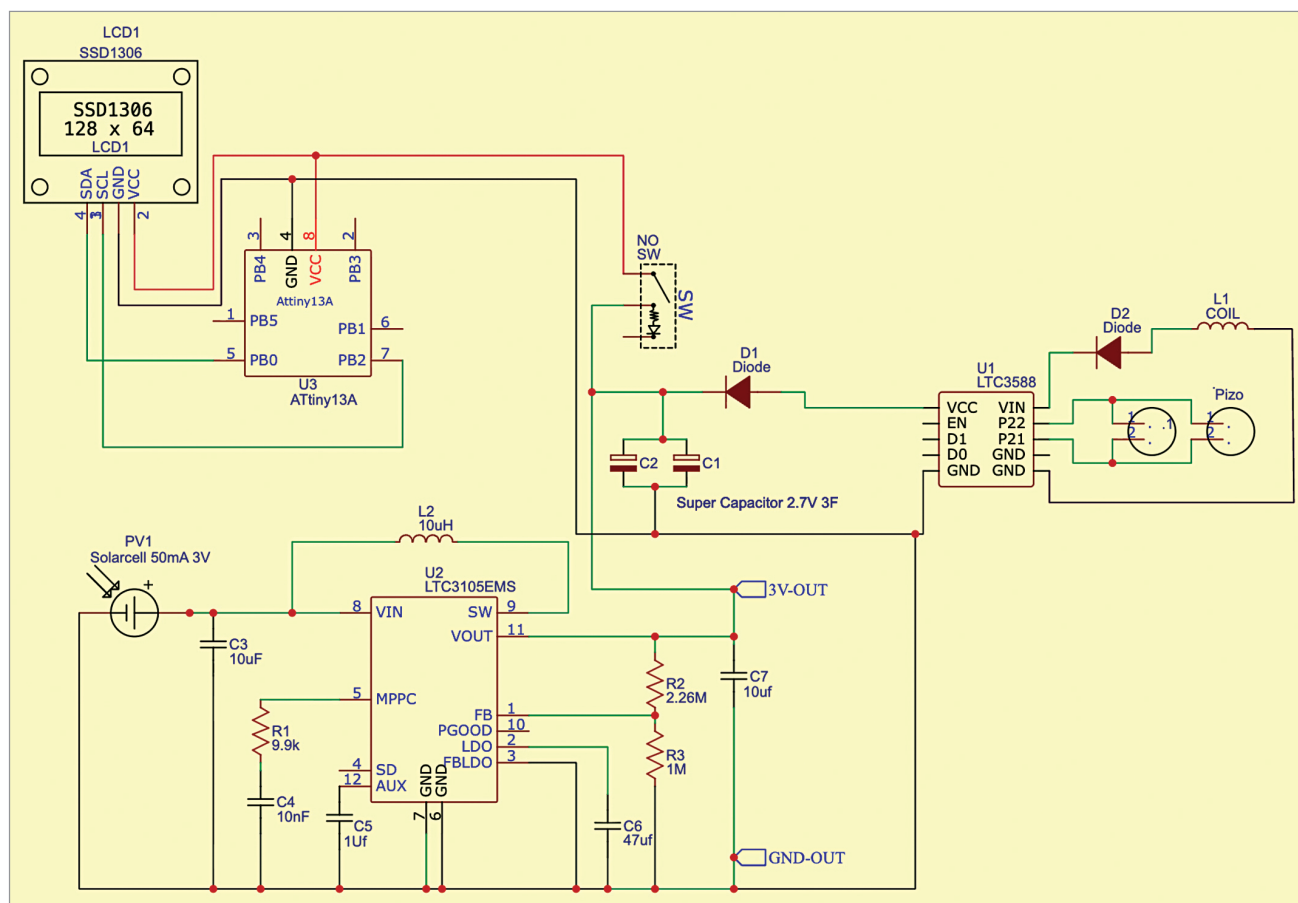


Fig. 2: Circuit diagram

measured power consumption.

Two approaches are possible.

The first connects the PV cell directly to the supercapacitor via a Schottky diode, enabling charge accumulation from ambient light. This can be combined with the LTC3588 energy-harvesting system for vibration or other mechanical sources, with both inputs feeding

the same supercapacitor.

The second approach uses two dedicated energy-harvesting ICs to create separate, optimised paths: one for solar energy and the other for vibration or electromagnetic sources. This method enables more efficient energy extraction from multiple ambient sources, with all harvested energy stored in a

common supercapacitor.

Table 3 shows the charging-time comparison between 10F and 20F supercapacitors. If we use supercapacitor, the charging time can be calculated as $t \text{ (seconds)} = C \times (V_{\text{final}} - V_{\text{initial}}) / I$

$C = 10\text{F}$ (capacitance)

$V_{\text{final}} = \approx 2.4\text{--}2.5\text{V}$ (typical with Schottky diode drop)